

# AAYUSH SALVI

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## EDUCATION

**University of California San Diego**  
Masters in Data Science

Present

**Atharva College of Engineering, Mumbai, India**  
Graduated with a Bachelor of Technology in Information Technology with *Distinction*  
Honors in Data Science

May 2024  
(CGPA 3.6/4)

## PROFESSIONAL EXPERIENCE

**Konsultera Solutions, Mumbai, India**  
*Software Developer*

April 2025 – August 2025

- Developed and deployed scalable API endpoints using FastAPI and FastMCP, implementing a custom Model Control Protocol (MCP) framework from scratch to orchestrate AI tool operations.
- Integrated LangChain pipelines to coordinate workflows between RAG and MCP systems, improving data retrieval relevance by 25%.
- Designed and optimized a production-grade asynchronous API system that handled 10K+ concurrent requests, reducing response latency by 40% through Python's AsyncIO.
- Containerized 5+ microservices with Docker and deployed on Google Cloud Run, achieving 95% uptime and ensuring smooth auto-scaling.
- Worked on a Context-Aware RAG implementation focused on improving the relevance of generated summaries.

**Coditation Systems, Pune, India**  
*Data Science Intern*

June 2023 – July 2023

- Conducted exploratory and trend analysis on business contract data using Microsoft Power BI to uncover patterns that guided forecasting and renewal strategies
- Performed data preprocessing and feature engineering on 50K+ contract records, creating new predictive variables for renewal and termination trends
- Built and evaluated time-series forecasting models (SARIMA) and ensemble classifiers, achieving an accuracy improvement of 12% over baseline models.

## ACADEMIC PROJECTS AND PAPER

### ByteBot



- Built an AI-based web platform using HTML, CSS, and JavaScript that enables non-technical users to train custom CNN models without prior ML experience.
- Designed and optimized a TensorFlow.js data pipeline, supporting datasets up to 2GB and achieving 35% faster model training through efficient client-side processing.
- Streamlined user experience by adding hyperparameter tuning, visualization, and model export modules, resulting in 50% higher model reuse rate among testers

### Gemstone Prediction



- Developed an end-to-end machine learning pipeline for gemstone price prediction using Python, sci-kit-learn, and Flask
- Achieved an  $R^2$  score of 0.98 and MAE of 628.87 using a Random Forest Regressor trained on 20K+ records
- Integrated MLOps best practices with DVC for version control and Apache Airflow for workflow orchestration, reducing retraining time by 60%
- Automated model monitoring and retraining, enabling reproducible and continuous delivery through CI/CD pipelines

### TV-Series Analysis using NLP and LLM



- Conducted NLP-driven analysis on scripts from 500+ TV episodes to identify key characters, relationships, and narrative themes.
- Extracted and visualized character networks using SpaCy's NER, NetworkX, and PyViz, improving interpretability of character interactions by 40%
- Developed a zero-shot theme classifier using BART-large-MNLI, achieving 92% F1-score on theme extraction tasks.
- Integrated all models into an interactive Gradio interface, delivering a seamless and engaging user experience with <2s inference latency.

## TECHNICAL SKILLS

- Programming Languages:** Python, R, SQL, Go
- Databases:** PostgreSQL, MySQL, MongoDB, Redis
- Libraries:** Pandas, NumPy, Sklearn, Keras, TensorFlow, Matplotlib, nltk, transformers, LangChain, Pytorch, FastAPI, FastMCP, git, Flask, HuggingFace
- General Skills:** Data Science, Machine Learning, Predictive Modelling, Data Analysis, Deep Learning, Statistical Analysis, Time-Series Analysis, GenAI, Prompt Engineering, API Development, NLP, LLM Applications, CI/CD Pipelines
- MLOps Foundation** – iNeuron Certification (July 2024)
- OCI Generative AI Professional** – Oracle Certification (June 2024)